# Week 1 Data Analysis and Python Assessment

IIT Gandhinagar | Cohort 2 | Saturday 10 October 2026

120 minutes | 100 marks | 17 questions

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Answer all questions on separate answer sheets. Show working. Write code by hand where requested; minor syntax slips will not lose marks when the logic is clear. No calculator, laptop, phone or notes. Statistical questions require interpretation, not calculation of test statistics.

Revenue means completed-order value after discounts and refunds. The small Kalpa scenarios and campaign estimates in this paper are illustrative exam data. Every question supplies the information it needs.

## Section A Business analysis and table relationships

31 marks | Suggested time 40 minutes

### Q1 Did retention improve or did the denominator change [8 marks]

Kalpa runs a paid membership programme. A member is due for renewal when their current membership expires during the reporting period.

| Measure | Previous year | Current year |
|---|---|---|
| Members due for renewal | 1,000 | 1,500 |
| Members who renewed | 800 | 1,050 |
| Revenue from those renewals | ₹16 lakh | ₹21 lakh |

The membership head says: "We retained 250 more members and earned ₹5 lakh more. Clearly, retention improved."

a. Calculate the renewal rate and revenue per renewed member for both years. [2]

b. Evaluate the statement. Explain how renewal numbers and revenue can increase while the renewal rate decreases. Use the figures. [2]

Further analysis produces this breakdown:

| Membership type | Previous due | Previous renewed | Current due | Current renewed |
|---|---|---|---|---|
| Monthly | 500 | 350 | 1,000 | 600 |
| Annual | 500 | 450 | 500 | 450 |

c. Calculate the renewal rate for each type in both years. Explain what the overall result hides about changes within a membership type and changes in the composition of members due for renewal. [2]

d. Before recommending a discount to every member due for renewal, identify one further comparison you would request. Explain how its result could change your recommendation. [2]

## Q2 Is a fall in average frequency enough to identify the problem [8 marks]

Kalpa compares two equal-length reporting periods.

| Customer group | Previous buyers | Previous orders | Current buyers | Current orders |
|---|---|---|---|---|
| Purchased in both periods | 600 | 3,000 | 600 | 3,000 |
| Previous period only | 400 | 2,000 | 0 | 0 |
| Current period only | 0 | 0 | 900 | 1,800 |
| Total | 1,000 | 5,000 | 1,500 | 4,800 |

The CEO says: "Customers increased, but orders per customer fell. Our existing customers must be purchasing less often."

a. Calculate overall orders per purchasing customer in both periods and for customers who purchased in both periods. [2]

b. Does the table support the CEO's explanation? Explain how buyers leaving and entering affect the overall average. [2]

c. A colleague says: "The returning group's average stayed unchanged, so none of those customers changed their behaviour." Is this justified? Construct an example of two returning customers whose combined order count stays unchanged while their individual behaviour changes. [2]

d. Describe the table and comparison you would create for the 600 returning customers to establish whether deterioration is widespread or concentrated. [2]

## Q3 Connect the tables before trusting the calculation [8 marks]

You receive these four tables. The description above each table states what one row represents.

Customers — one row per registered customer

| customer_id | segment |
|---|---|
| C01 | Plus |
| C02 | Core |
| C03 | Plus |

Orders — one row per order

| order_id | customer_id | period_id | status |
|---|---|---|---|
| O11 | C01 | P1 | COMPLETED |
| O12 | C01 | P2 | COMPLETED |
| O13 | C02 | P2 | CANCELLED |
| O14 | C03 | P2 | COMPLETED |

Order items — one row per product line within an order

| order_id | line_number | product_id | net_amount |
|---|---|---|---|
| O11 | 1 | A | 400 |
| O11 | 2 | B | 600 |
| O12 | 1 | A | 700 |
| O13 | 1 | C | 500 |
| O14 | 1 | B | 300 |
| O14 | 2 | B | 200 |

Periods — one row per reporting period

| period_id | period_label |
|---|---|
| P1 | Previous |
| P2 | Current |

a. Identify a suitable unique key for each table. A key may contain multiple columns. Explain why product_id cannot uniquely identify an order-item row. [2]

b. State which columns connect the tables. Describe the relationship between customers and orders, and between orders and order items. [2]

c. An analyst connects orders to order items and counts the resulting rows as orders. Using O11 and O14, explain the mistake. State the correct number of completed orders in each period. [2]

d. Calculate current-period completed-order revenue for each segment. Explain your treatment of the cancelled order and the two lines belonging to O14. [2]

## Q4 Which campaign deserves more money [7 marks]

These are illustrative campaign estimates. Contribution before advertising is revenue remaining after discounts and order-related costs, before advertising cost.

| Measure | Campaign A | Campaign B |
|---|---|---|
| Advertising spend | ₹10 lakh | ₹10 lakh |
| Revenue attributed to campaign | ₹50 lakh | ₹35 lakh |
| Contribution before advertising | ₹7.5 lakh | ₹14 lakh |

ROAS = revenue attributed to advertising ÷ advertising spend.

a. Calculate ROAS and contribution after advertising for both campaigns. [2]

b. Marketing prefers Campaign A because its ROAS is higher. Evaluate this preference using the figures. [2]

c. Why does attributed revenue not necessarily mean revenue caused by the campaign? Describe a comparison that would help investigate the difference. [1]

d. Before spending, propose one measurable result for deciding whether to continue and one specific condition that would make you pause or stop. [2]

## Section B Python reasoning

25 marks | Suggested time 25 minutes

### Q5 Two names and one list [5 marks]

```
amounts = [100, 200, 300]
first = amounts
second = amounts[:]
first[1] = 900
second.append(400)
print(amounts)
print(first)
print(second)
```

a. Write the three outputs. [3]

b. Explain why changing first affects amounts, while appending to second does not. [2]

### Q6 Clean a customer code [5 marks]

Write a function clean_customer_code that removes surrounding spaces, converts letters to uppercase, removes the hyphen and returns the cleaned code. Input " c-007 " must return "C007".

a. Write the function. [3]

b. Explain what would change if the final line used print instead of return and someone ran the following statement. [2]

```
code = clean_customer_code(" c-007 ")
```

## Q7 A condition that looks right [5 marks]

```
orders = [
    {"id": "O1", "status": "COMPLETED", "amount": 500},
    {"id": "O2", "status": "CANCELLED", "amount": 700},
    {"id": "O3", "status": "COMPLETED", "amount": 300}
]
total = 0
for order in orders:
    if order["status"] == "COMPLETED" or "REFUNDED":
        total += order["amount"]
print(total)
```

a. What does the code print? Explain why. [2]

b. Correct the condition to accept only COMPLETED or REFUNDED orders. [1]

c. For Kalpa's completed-order analysis, write the condition that should actually be used and state the resulting total. [2]

## Q8 One bad record should not end the analysis [5 marks]

```
records = [
    {"id": "A1", "amount": "1200"},
    {"id": "A2", "amount": "invalid"},
    {"id": "A3"},
    {"id": "A4", "amount": "800"}
]
total = 0
for record in records:
    total += int(record["amount"])
print(total)
```

a. Identify the first record that causes an error and name the error type. [1]

b. Rewrite the loop using try and except to add readable amounts, print IDs for invalid or missing amounts, and continue processing. [3]

c. State the final total after your correction. [1]

## Q9 Does this count customers or orders [5 marks]

```
orders = [
    {"customer_id": "C1", "amount": 100},
    {"customer_id": "C2", "amount": 200},
    {"customer_id": "C1", "amount": 300},
    {"customer_id": "C3", "amount": 400},
    {"customer_id": "C2", "amount": 500}
]
counts = {}
for order in orders:
    customer = order["customer_id"]
    if customer not in counts:
        counts[customer] = 0
    counts[customer] += 1
```

a. Write the final contents of counts. *[2]*

b. Using counts, write code to calculate total orders and distinct purchasing customers. *[2]*

c. State orders per purchasing customer; a fraction is acceptable. *[1]*

## Section C Pandas indexing and updating

20 marks | Suggested time 20 minutes

Use this original DataFrame independently for each of Questions 10 to 13.

```
import pandas as pd
df = pd.DataFrame(
    {
        "customer_id": ["C1", "C2", "C3", "C4", "C5"],
        "segment": ["Core", "Plus", "Plus", "Core", "Plus"],
        "previous_orders": [5, 6, 0, 4, 3],
        "current_orders": [4, 2, 3, 4, 0]
    },
    index=[10, 20, 30, 40, 50]
)
```

| Index | customer_id | segment | previous_orders | current_orders |
|---|---|---|---|---|
| 10 | C1 | Core | 5 | 4 |
| 20 | C2 | Plus | 6 | 2 |
| 30 | C3 | Plus | 0 | 3 |
| 40 | C4 | Core | 4 | 4 |
| 50 | C5 | Plus | 3 | 0 |

### Q10 Labels and positions [5 marks]

a. Write the rows or values returned by each expression. You need not reproduce pandas formatting. *[4]*

```
df.loc[20, "current_orders"]
df.iloc[1, 3]
df.loc[20:40, ["customer_id", "current_orders"]]
df.iloc[1:3, [0, 3]]
```

b. Explain why the last two expressions return different sets of rows. [1]

### Q11 Select the customers we need [5 marks]

a. Write pandas code selecting customers who purchased in both years but placed fewer current-year orders. Show only customer_id, previous_orders and current_orders. [3]

b. Select Plus customers who purchased in the previous year but placed no orders in the current year. [2]

### Q12 Correct a record and create a column [5 marks]

a. C2 placed 3, not 2, current-year orders. Write one statement updating the correct cell. [2]

b. Create order_change as current_orders minus previous_orders. [1]

c. Create a Boolean declined column, True only when the customer had previous-year orders and placed fewer current-year orders. [2]

### Q13 Find the mistake [5 marks]

```
df.loc[
   df["segment"] == "Plus" and df["current_orders"] < 3,
   "customer_id"
]
```

a. Explain why the condition is incorrect for pandas. [2]

b. Write the corrected statement. [2]

c. List the selected customer IDs from the original DataFrame. [1]

## Section D Statistical understanding

24 marks | Suggested time 35 minutes

Do not calculate z-statistics, p-values, standard deviations or confidence intervals.

### Q14 Which distribution are we discussing [8 marks]

Kalpa's individual order values are strongly right-skewed. An analyst independently draws 500 random samples, each containing 100 orders. She calculates the mean of each sample and plots the 500 means.

a. Distinguish the distribution of individual order values from the distribution of the 500 sample means. [2]

b. Explain what the central limit theorem suggests about the second distribution, assuming independent observations, finite variance and an adequate sample size. Does it imply that individual order values become normally distributed? [2]

c. She repeats the exercise using 400 orders per sample. What should happen to the centre and spread of the sample means? Explain without calculating. [2]

d. Another analyst samples only app orders. He says: "My samples are large and the sample means look normal. Therefore, my estimate represents all Kalpa orders." Explain why this is unjustified. [2]

## Q15 Is being near the mean always desirable [6 marks]

| Situation | Reference mean | Observation |
|---|---|---|
| Delivery time | 40 minutes | A delivery takes 20 minutes |
| Customer contribution | ₹500 per customer | A customer contributes ₹1,500 |
| Package weight; labelled target 1,000 g | 1,000 g | A package weighs 900 g |

a. For each situation, explain whether being below, near or above the mean is desirable. State any additional information needed before judging performance. [3]

b. A colleague says: "An observation far from the mean is an outlier, so remove it before analysis." Explain why this is unsafe, using one situation above. [2]

c. Mean delivery time rises from 40 to 60 minutes. Does a 60-minute delivery becoming average make it acceptable? Explain. [1]

## Q16 Interpret a z-test [5 marks]

Kalpa treats 20% as the maximum acceptable non-return rate for the wider customer population being studied. In an appropriate random sample of 400 previous-year buyers, 121 did not purchase again.

For a one-sided z-test of whether the population rate exceeds 20%, the analyst reports $z = 5.125$ and p-value approximately 0.00000015. Assume sampling and approximation requirements are satisfied. The decision threshold was set at 0.05 before analysis.

a. State the null and alternative hypotheses in words. [2]

b. Interpret the p-value in this situation. [1]

c. What conclusion is supported at the stated threshold? [1]

d. Does the result establish that reducing marketing spending would solve the problem? Explain. [1]

## Q17 No convincing difference does not mean identical behaviour [5 marks]

| Group | Previous-year buyers | Did not return | Non-return rate |
|---|---|---|---|
| Previously frequent buyers | 3,926 | 1,153 | 29.37% |
| Other previous-year buyers | 6,074 | 1,847 | 30.41% |

Previously frequent means at least five completed orders in the previous year. A statistical comparison reports a two-sided p-value of 0.268. Assume the inferential comparison and its requirements are appropriate; use a threshold of 0.05.

a. A colleague says: "The test proves the groups behave identically." Explain why this is incorrect. [2]

b. Write a more accurate conclusion in two sentences, referring to the observed rates and the test result. [2]

c. Another colleague says: "Neither group requires attention because the test was not significant." Explain why that does not follow. [1]

End of paper